

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

**(Permanent Affiliation to JNTUK, Kakinada),
PEDATADEPALLI, TADEPALLIGUDEM-534 101**

A.Y: 2024-25

IV SEM CSE Handbook (V23 Regulation)



Department of Computer Science and Engineering (Accredited by NBA)

Pedatadepalli, Tadepalligudem-534101, A.P

INDEX

S.NO	CONTENTS	PAGE NO.
1.	Institute Vision & Mission	4
2.	Department Vision & Mission	6
3.	Program Educational Objectives, Program Outcomes & Program Specific Outcomes	8
4.	Academic Calendar	10
5.	Class Time Table	11
6.	Course Structure	13
Lesson Plan		
7.	Universal Human Values (Lesson Plan)	15
8.	Probability and Statistics (Lesson Plan)	20
9.	Operating Systems (Lesson Plan)	27
10.	Database Management Systems (Lesson Plan)	33
11.	Software Engineering (Lesson Plan)	39
12.	Operating Systems Lab (Lesson Plan)	45
13.	Database Management Systems Lab (Lesson Plan)	49
14.	Full Stack Development –I (Lesson Plan)	52
15.	Environmental Science (Lesson Plan)	56
16.	Professional Communication Skills –II (Lesson Plan)	63



INSTITUTE

VISION MISSION



INSTITUTE VISION & MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.



DEPARTMENT

VISION

MISSION

DEPARTMENT VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

**PROGRAM
EDUCATIONAL
OBJECTIVES,
PROGRAM OUTCOMES
& PROGRAM
SPECIFIC
OUTCOMES**

Programme Educational Objectives (PEOs)

Graduates of this programme will :

- PEO 1:** Adapt to evolving technology.
- PEO 2:** Provide optimal solutions to real time problems.
- PEO 3:** Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

- PSO 1:** Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. **[K3]**
- PSO 2:** Use Professional engineering practices and strategies for development and maintenance of software. **[K3]**

Program Outcomes (POs): Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. **[K3]**
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

✉ : principal@srivasaviengg.ac.in
svec.a8@gmail.com



☎ : 08818- 284344, 355

SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)
(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)
(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12(B))
(NBA Accreditation to B.Tech., EEE, CSE, ME and ECE Branches for 3 Years)
Pedatadepalli, **TADEPALLIGUDEM** – 534 101. W.G.Dist. (A.P)

Principal's Office
Date: 03-07-2024

Academic Calendar **For II B.Tech (III and IV Semesters), Academic Year 2024-25**

III Semester			
Description	From	To	Weeks
Commencement of Class Work	22.07.2024		
I Unit of Instructions	22.07.2024	14.09.2024	8 W
I Mid Examinations	16.09.2024	21.09.2024	1 W
II Unit of Instructions	23.09.2024	16.11.2024	8 W
II Mid Examinations	18.11.2024	23.11.2024	1 W
Preparation & Practicals	25.11.2024	30.11.2024	1 W
End Examinations	02.12.2024	14.12.2024	2 W
Commencement of Next Semester Class Work (IV Semester)	16.12.2024		
IV Semester			
I Unit of Instructions	16.12.2024	08.02.2025	8 W
I Mid Examinations	10.02.2025	15.02.2025	1 W
II Unit of Instructions	17.02.2025	12.04.2025	8 W
II Mid Examinations	14.04.2025	19.04.2025	1 W
Preparation & Practicals	21.04.2025	26.04.2025	1 W
End Examinations	28.04.2025	10.05.2025	2 W
Community Service Project	12.05.2025	05.07.2025	8 W
Commencement of Next Semester Class Work (V Semester)	07.07.2025		


PRINCIPAL

Copy to : ALL

Vision

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

Mission

- To produce Engineering graduates of professional quality and global perspective through learner-centric education.
- To establish linkages with government, industry and Research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



IV SEM CLASS CONSOLIDATED TIMETABLE

w.e.f.20-12-2024

Section: A

Class Coordinator: Mrs. M.N.V. Surekha

Room No : B-301

Periods	1	2	3	4	1:00PM - 2:00PM	5	6	7
Time	(09.30 AM - 10.30 AM)	(10.30 AM - 11.20 AM)	(11.20 AM - 12.10 PM)	(12.10 PM - 01.00 PM)		(02.00 PM - 02.50 PM)	(02.50 PM - 03.40 PM)	(03.40 PM - 04.30 PM)
Day								
Mon	DBMS	P&S	PCS-II (Aptitude)		Lunch Break	SE	OS	UHV
Tue	OS	P&S	ES	SE		UHV	P&S	DBMS
Wed	P&S	DBMS	UHV	OS		OS Lab		
Thu	FSD-I	FSD-I Lab				SE	DBMS	UHV
Fri	SE	DBMS Lab				ES	OS	Library
Sat	P&S	OS	SE	DBMS		PCS-II (English)		Sports

Section: B

Class Coordinator: Dr. G Tej Varma

Room No : B-302

Periods	1	2	3	4	1:00PM - 2:00PM	5	6	7
Time	(09.30 AM - 10.30 AM)	(10.30 AM - 11.20 AM)	(11.20 AM - 12.10 PM)	(12.10 PM - 01.00 PM)		(02.00 PM - 02.50 PM)	(02.50 PM - 03.40 PM)	(03.40 PM - 04.30 PM)
Day								
Mon	OS	OS Lab			Lunch Break	DBMS	PCS-II (Aptitude)	
Tue	DBMS	DBMS Lab				OS	P&S	ES
Wed	SE	OS	P&S	DBMS		FSD-I Lab		
Thu	DBMS	SE	PCS-II (English)			UHV	OS	SE
Fri	FSD-I	P&S	UHV	SE		UHV	DBMS	Library
Sat	P&S	ES	UHV	OS		SE	P&S	Sports

Section: C

Class Coordinator: Mrs. A Naga Jyothi

Room No : B-303

Periods	1	2	3	4	1:00PM - 2:00PM	5	6	7
Time	(09.30 AM - 10.30 AM)	(10.30 AM - 11.20 AM)	(11.20 AM - 12.10 PM)	(12.10 PM - 01.00 PM)		(02.00 PM - 02.50 PM)	(02.50 PM - 03.40 PM)	(03.40 PM - 04.30 PM)
Day								
Mon	DBMS	PCS-II (English)		SE	Lunch Break	P&S	OS	UHV
Tue	SE	FSD-I	OS	P&S		DBMS	PCS-II (Aptitude)	
Wed	P&S	DBMS Lab				OS	ES	SE
Thu	OS	DBMS	P&S	UHV		FSD-I Lab		
Fri	SE	OS Lab				DBMS	UHV	Library
Sat	OS	SE	DBMS	P&S		UHV	ES	Sports

Section: D

Class Coordinator: M V V Gopala Krishna Murthy

Room No : B-304

Period s	1	2	3	4	1:00PM - 2:00PM	5	6	7
Time	(09.30 AM - 10.30 AM)	(10.30 AM - 11.20 AM)	(11.20 AM - 12.10 PM)	(12.10 PM - 01.00 PM)		(02.00 PM - 02.50 PM)	(02.50 PM - 03.40 PM)	(03.40 PM - 04.30 PM)
Day								
Mon	OS	SE	DBMS	OS	Lunch Break	DBMS Lab		
Tue	P&S	DBMS	PCS-II (Aptitude)			OS Lab		
Wed	OS	FSD-I	P&S	OS		UHV	SE	Library
Thu	SE	UHV	OS	DBMS		P&S	PCS-II (English)	
Fri	ES	P&S	SE	UHV		FSD-I Lab		
Sat	DBMS	P&S	UHV	SE		DBMS	ES	Sports

Staff Details:

S. No.	Course Code	Course Name	Name of the Faculty			
			Section 1	Section 2	Section 3	Section 4
1.	V23MBT53	Universal Human Values (UHV)	Mrs. N. Hiranmayee	Mrs. N. Hiranmayee	Ms. Y Divya Vani	Ms. Ch Naga Padma Latha
2.	V23MAT07	Probability and Statistics (P&S)	Mr. T D Rama Krishna	Mrs. S Sirisha	Ms.P Sravani	Ms.T.S Suryaprabha
3.	V23CST06	Operating Systems (OS)	Dr. G Tej Varma	Dr. G Tej Varma	Mrs. A Naga Jyothi	Mrs. A Naga Jyothi
4.	V23CST07	Database Management Systems (DBMS)	Mrs. M.N.V.Surekha	Mrs. M.N.V.Surekha	M V V Gopala Krishna Murthy	M V V Gopala Krishna Murthy
5.	V23CST08	Software Engineering (SE)	Mrs. G. Prasanthi	Mrs. G. Prasanthi	Mr. G.Sriram Ganesh	Mr. G.Sriram Ganesh
6.	V23CSL06	Operating Systems Lab (OS LAB)	Dr. G Tej Varma / Mrs. N Hiranmayee	Dr. G Tej Varma / Ms. Y Divya Vani	Mrs. A Naga Jyothi / Ms. R Nava Lavanya	Mrs. A Naga Jyothi / Mr. M.V.V. Krishna
7.	V23CSL07	Database Management Systems Lab (DBMS LAB)	Mrs.M.N.V.Surekha / Mrs.S. Santhi Rupa	Mrs. M.N.V. Surekha / Mr.Sd Sadik	M V V Gopala Krishna Murthy / Mr. P. Rama Mohan Rao	M V V Gopala Krishna Murthy / Mr. P. Rama Mohan Rao
8.	V23CSSE02	Full Stack Development –I (FSD-I)	Dr. D. Jaya Kumari / Mr. T Anil Kumar Reddy	Dr. D. Jaya Kumari / Mr. T Anil Kumar Reddy	Mr. T Anil Kumar Reddy / Ms. R Nava Lavanya	Mr. T Anil Kumar Reddy / Ms. R Nava Lavanya
9.	V23CEAC01	Environmental Science (ES)	Mrs.S.B.Sarojini	Mrs.L Jyotsna	Mrs.L Jyotsna	Ms. Md. Syed Yasmin
10.	V23ENT02	Professional Communication Skills –II (PCS-II) - Aptitude	Mr.J.N.V.Somayajulu	Mr.T.H.S.Srinivas	Mr.J.N.V.Somayajulu	Mr.T.H.S.Srinivas
		Professional Communication Skills –II (PCS-II) - English	Mr G Srinivasa Rao	Mr.M. Venkata Ramana	Dr K Venkata Rao	Mrs. Tanuja.Ch

Lab Venues:

S. No.	Name of the Lab Course	Lab Venue
1	Operating Systems Lab (OS LAB)	PG CP Lab
2	Database Management Systems Lab (DBMS LAB)	James Gosling Lab
3	Full Stack Development –I (FSD-I)	



Head of the Department

COURSE STRUCTURE

SEMESTER - IV (SECOND YEAR)

S. No.	Course Code	Name of the Course		L	T	P	C
1	V23MBT53	Universal Human Values	BS&H	2	1	0	3
2	V23MATO7	Probability & Statistics	BS&H	3	0	0	3
3	V23CSTO6	Operating Systems	PCC	3	0	0	3
4	V23CSTO7	Database Management Systems	PCC	3	0	0	3
5	V23CSTO8	Software Engineering	PCC	2	1	0	3
6	V23CSLO6	Operating Systems Lab	PCC	0	0	3	1.5
7	V23CSLO7	Database Management Systems Lab	PCC	0	0	3	1.5
8	V23CSSE02	Full Stack Development-I	Skill Enhancement Course	0	1	2	2
9	V23CEACO1	Environmental Science	Audit Course	2	0	0	-
10	V23ENTO2	Professional Communication Skills-II	Audit Course	2	0	0	0
Total				17	3	8	20
Mandatory Community Service Project (V23COSP01) Internship of 08 weeks duration during summer vacation							

**LESSON
PLANS**

Universal Human Values

Academic Year: 2024-25

Year/ Semester: IV

Name of the Course: Universal Human Values

Programme: B.Tech

Section: A,B,C & D

Course Code: V23MBT53

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1	CO 1	Understand the concepts like Natural Acceptance, Happiness and Prosperity.	K2
2	CO2	Identify one's self, and one's surroundings (family, society nature).	K2
3	CO 3	Apply what they have learnt to their own self in different day-to-day settings in real life.	K3
4	CO4	Relate human values with human relationship and human society.	K3
5	CO 5	Discover as socially and ecologically responsible engineers.	K3

Text Books:

1. R R Gaur, R Asthana, G P Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1

Reference Books

1. JeevanVidya: EkParichaya, A Nagaraj, JeevanVidya Prakashan, Amarkantak, 1999.
2. Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. The Story of Stuff (Book).
4. The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
5. Small is Beautiful - E. F Schumacher.
6. Slow is Beautiful - Cecile Andrews
7. Economy of Permanence - J C Kumarappa explore
8. Bharat Mein Angreji Raj – PanditSunderlal

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level (Marks In %)		60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
UNIT I: Introduction to Value Education (6 lectures and 3 tutorials for practice session)						
1	CO-1	Introduction to OBE, Dissemination of Vision, Mission of the Dept. and PEOs, POs& PSOs of the Programme.		1	Lecture	BB
2		Course Introduction- Need, Basic Guidelines, Content & Process of Value Education.	K1	1	Lecture	BB
3		Lecture 1: Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education).	K1	1	Lecture	BB
4		Lecture 2: Understanding Value Education.	K2	2	Lecture with discussion	BB
5		Tutorial 1: Practice Session PS1 Sharing about Oneself	K2	1	Lecture	BB+ICT
6		Lecture 3: self-exploration as the Process for Value Education.	K1	1	Lecture	BB+ICT
7		Lecture4: Continuous Happiness and Prosperity – the Basic Human Aspirations	K2	2	Lecture with Discussion	BB+ICT
8		Tutorial 2: Practice Session PS2 Exploring Human Consciousness.	K2	1	Lecture	BB+ICT
9		Lecture 5: Happiness and Prosperity – Current Scenario.	K1	1	Lecture	BB
10		Lecture 6: Method to Fulfill the Basic Human Aspirations	K1	1	Lecture	BB
11		Tutorial 3: Practice Session PS3 Exploring Natural Acceptance	K2	1	Lecture	BB
UNIT II: Harmony in the Human Being (6 lectures and 3 tutorials for practice session)						
12	CO-2	Lecture 7: Understanding Human being as the Co-existence of the self and the body.	K2	1	Lecture with discussion	BB+ICT

13		Lecture 8: Distinguishing between the Needs of the self and the body	K2	1	Lecture with discussion	BB+ICT
14		Tutorial 4: Practice Session PS4 Exploring the difference of Needs of self and body.	K2	1	Lecture with discussion	BB+ICT
15		Lecture 9: The body as an Instrument of the self	K2	1	Lecture with discussion	BB+ICT
16		Lecture 10: Understanding Harmony in the self	K2	1	Lecture with discussion	BB+ICT
17		Tutorial 5: Practice Session PS5 Exploring Sources of Imagination in the self	K2	1	Lecture with discussion	BB+ICT
18		Lecture 11: Harmony of the self with the body	K2	1	Lecture with discussion	BB+ICT
19		Lecture 12: Programme to ensure self-regulation and Health	K2	1	Lecture with discussion	BB+ICT
20		Tutorial 6: Practice Session PS6 Exploring Harmony of self with the body	K2	1	Lecture with discussion	BB+ICT

UNIT III: Harmony in the Family and Society (6 lectures and 3 tutorials for practice session)

21	CO-3	Lecture 13: Harmony in the Family – the Basic Unit of Human Interaction	K3	1	Lecture with discussion	BB+ICT
22		Lecture 14: 'Trust' – the Foundational Value in Relationship	K3	1	Lecture with discussion	BB+ICT
23		Tutorial 7: Practice Session PS7 Exploring the Feeling of Trust	K3	1	Lecture with discussion	BB+ICT
24		Lecture 15: 'Respect' – as the Right Evaluation	K3	1	Lecture with discussion	BB+ICT
25		Tutorial 8: Practice Session PS8 Exploring the Feeling of Respect	K3	1	Lecture with discussion	BB+ICT
26		Lecture 16: Other Feelings, Justice in Human-to-Human Relationship	K3	1	Lecture with discussion	BB+ICT
27		Lecture 17: Understanding Harmony in the Society	K2	1	Lecture with discussion	BB+ICT

28		Lecture 18: Vision for the Universal Human Order	K3	1	Lecture with discussion	BB+ICT
29		Tutorial 9: Practice Session PS9 Exploring Systems to fulfil Human Goal	K3	1	Lecture with discussion	BB+ICT
UNIT IV: Harmony in the Nature/Existence (4 lectures and 2 tutorials for practicesession)						
30	CO-4	Lecture 19: Understanding Harmony in the Nature	K3	1	Lecture with discussion	BB+ICT
31		Lecture 20: Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature	K3	1	Lecture with discussion	BB+ICT
32		Tutorial 10: Practice Session PS10 Exploring the Four Orders of Nature	K3	1	Lecture with discussion	BB+ICT
33		Lecture 21: Realizing Existence as Co-existence at All Levels	K3	1	Lecture with discussion	BB+ICT
34		Lecture 22: The Holistic Perception of Harmony in Existence	K3	1	Lecture with discussion	BB+ICT
35		Tutorial 11: Practice Session PS11 Exploring Co-existence in Existence.	K3	1	Lecture with discussion	BB+ICT
UNIT V: Implications of the Holistic Understanding – a Look at Professional Ethics (6 lectures and 3 tutorials for practice session)						
36	CO-5	Lecture 23: Natural Acceptance of Human Values	K3	1	Lecture with discussion	BB+ICT
37		Lecture 24: Definitiveness of (Ethical) Human Conduct	K3	1	Lecture with discussion	BB+ICT
38		Tutorial 12: Practice Session PS12 Exploring Ethical Human Conduct	K3	1	Lecture with discussion	BB+ICT

39		Lecture 25: A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order	K3	1	Lecture with discussion	BB+ICT
40		Lecture 26: Competence in Professional Ethics	K3	1	Lecture with discussion	BB+ICT
41		Tutorial 13: Practice Session PS13 Exploring Humanistic Models in Education	K3	1	Lecture with discussion	BB+ICT
42		Lecture 27: Holistic Technologies, Production Systems and Management Models-Typical Case Studies	K3	1	Lecture with discussion	BB+ICT
43		Lecture 28: Strategies for Transition towards Value-based Life and Profession	K3	1	Lecture with discussion	BB+ICT
44		Tutorial 14: Practice Session PS14 Exploring Steps of Transition towards Universal Human Order	K3	1	Lecture with discussion	BB+ICT

Total No. of Classes: 44

Probability & Statistics

Academic Year: 2024-25

Programme: B.Tech

Year/ Semester: IV

Section: A,B,C & D

Name of the Course: Probability & Statistics

Course code: V23MAT07

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S . No.	CO No.	Course Outcome	BTL
1	CO1	Classify the concepts of data science and its importance .	K2
2	CO2	Interpret the association of characteristics and through correlation and regression tools.	K3
3	CO3	Apply discrete and continuous probability distributions.	K3
4	CO4	Construct the components of a classical hypothesis test.	K3
5	CO5	Apply the statistical inferential methods to test the hypothesis based on small and large samplings.	K3

Text Books:

1. **Miller and Freund's**, Probability and Statistics for Engineers, 7/e, Pearson, 2008.
2. **S. C. Gupta and V.K. Kapoor**, Fundamentals of Mathematical Statistics, 11/e, Sultan Chand & Sons Publications, 2012.

Reference Books:

1. **Shron L. Myers, Keying Ye, Ronald E Walpole**, Probability and Statistics Engineers and the Scientists, 8th Edition, Pearson 2007.
2. **Jay I. Devore**, Probability and Statistics for Engineering and the Sciences, 8th Edition, Cengage.
3. **Sheldon M. Ross**, Introduction to probability and statistics Engineers and the Scientists, 4th Edition, Academic Foundation, 2011.
4. **Johannes Ledolter and Robert V. Hogg**, Applied statistics for Engineers and Physical Scientists, 3rd Edition, Pearson, 2010.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level (Marks In %)		60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan:

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
Unit – 1: Descriptive statistics and methods for data science			K2	12		
1	CO1	Understand the role of statistics in data science and differentiate between population and sample.	K1	1	Lecture	Black Board
2	CO1	Identify different data collection methods	K1	1	Lecture	Black Board
3	CO1	Recognise the differences between primary and secondary data.	K2	1	Lecture with Discussion	Black Board
4	CO1	Classify variables as categorical or continuous.	K2	1	Lecture with Discussion	Black Board
5	CO1	Interpret patterns and insights from charts and graphs.	K2	1	Lecture with Discussion	Black Board
6	CO1	Interpret measures of central tendency.	K2	1	Lecture with Discussion	Black Board
7	CO1	Mean, Median, and Mode: Definitions and interpretations.	K2	1	Lecture with Discussion	Black Board
8	CO1	Explain Range, Variance, Standard Deviation	K1	1	Lecture	Black Board
9	CO1	Interpret Range, Variance, Standard Deviation	K2	1	Lecture with Discussion	Black Board
10	CO1	Interpret the skewness of a dataset.	K2	1	Lecture with Discussion	Black Board
11	CO1	Definition of kurtosis: leptokurtic, mesokurtic, and platykurtic distributions	K1	1	Lecture	Black Board
12	CO1	Recognise the shape of the data distribution by kurtosis	K2	1	Lecture with Discussion	Black Board

UNIT-II: Correlation and Regression			K3	10		
13	CO2	Define correlation and types of correlation, correlation coefficient	K1	1	Lecture	Black Board
14	CO2	Calculate Karl Pearson's coefficient of correlation for given data	K3	1	Lecture with Discussion and in class Assignment	Black Board
15	CO2	Discuss Spearman's rank correlation coefficient	K2	1	Lecture with Discussion	Black Board
16	CO2	Find the rank correlation coefficient	K3	1	Lecture with Discussion and in class Assignment	Black Board
17	CO2	Discuss regression and regression coefficients	K2	1	Lecture with Discussion	Black Board
18	CO2	Calculate the regression equation for given data	K3	1	Lecture with Discussion and in class Assignment	Black Board
19	CO2	Explain Least- Squares method to construct a curve to the given data	K2	1	Lecture with Discussion	Black Board
20	CO2	Apply the Method of Least-Squares to construct a straight line for the given data	K3	1	Lecture with Discussion and in class Assignment	Black Board
21	CO2	Apply the Method of Least-Squares to construct a second degree curve for the given data	K3	1	Lecture with Discussion and in class Assignment	Black Board
22	CO2	Apply the Method of Least-Squares to construct an exponential and power curve for the given data	K3	1	Lecture with Discussion and in class Assignment	Black Board

UNIT–III: Probability and Distributions			K3	15		
23	CO3	Define Probability and basic definitions related to Probability.	K1	1	Lecture	Black Board
24	CO3	Explain Addition Theorem of Probability, Conditional Probability and Multiplication theorem	K2	1	Lecture with Discussion	Black Board
25	CO3	Explain Baye's Theorem on Probability and related problems.	K2	1	Lecture with Discussion	Black Board
26	CO3	Define random variables: discrete and continuous with examples	K1	1	Lecture	Black Board
27	CO3	Explain discrete probability distribution, probability function, density function and their properties, expectation, mean, variance, standard deviation of a probability distribution	K2	1	Lecture with Discussion	Black Board
28	CO3	Calculate expectation, mean, variance and standard deviation of discrete distribution	K3	1	Lecture with Discussion and in class Assignment	Black Board
29	CO3	Explain continuous probability distribution, probability function, density function and their properties	K2	1	Lecture with Discussion	Black Board
30	CO3	Find the mean, median, mode, variance and standard deviation of continuous probability distribution	K3	1	Lecture with Discussion and in class Assignment	Black Board
31	CO3	Explain Binomial distribution and its properties	K2	1	Lecture with Discussion	Black Board
32	CO3	Find the probability using Binomial distribution	K3	1	Lecture with Discussion and in class Assignment	Black Board
33	CO3	Explain Poisson distribution and its properties	K2	1	Lecture with Discussion	Black Board
34	CO3	Find the probability using Poisson	K3	1	Lecture	Black

		distribution			with Discussion and in class Assignment	Board
35	CO3	Explain Normal distribution and its properties	K2	1	Lecture with Discussion	Black Board
36	CO3	Find the probabilities by using Normal distribution	K3	1	Lecture with Discussion and in class Assignment	Black Board
37	CO3	Explain Exponential distribution and uniform distribution	K2	1	Lecture with Discussion	Black Board
UNIT-IV: Sampling Theory			K3	10		
38	CO4	Define population and samples, sampling theory	K1	1	Lecture	Black Board
39	CO4	Define sampling distribution of means (σ known)	K1	1	Lecture	Black Board
40	CO4	Calculate mean, variance and standard deviations of sampling distributions of means	K3	1	Lecture with Discussion and in class Assignment	Black Board
41	CO4	Define sampling distribution of mean (σ unknown)	K1	1	Lecture	Black Board
42	CO4	Explain Estimation, criteria of a good estimator, point and interval estimation	K2	1	Lecture with Discussion	Black Board
43	CO4	Define the maximum error and confidence interval for the mean of a populations	K1	1	Lecture	Black Board
44	CO4	Estimate the maximum error and the confidence interval for the mean of a population using various distributions	K2	1	Lecture with Discussion	Black Board
45	CO4	Explain Central Limit theorem (without proof) and related problems	K1	1	Lecture	Black Board
46	CO4	Explain t- distribution, F-distribution and chi-square distribution and related problems	K1	1	Lecture	Black Board

47	CO4	Estimate the confidence interval for the mean of a population using t-distribution, F-distribution and chi-square distributions	K2	1	Lecture with Discussion	Black Board
UNIT–V: Tests of Hypothesis			K3	13		
48	CO5	Define Null and Alternative Hypothesis	K1	1	Lecture	Black Board
49	CO5	Explain Type-I, Type II errors and one tail and two tail test	K2	1	Lecture with Discussion	Black Board
50	CO5	Examine the hypothesis concerning mean using z-test	K3	1	Lecture with Discussion and in class Assignment	Black Board
51	CO5	Examine the hypothesis concerning proportion using z-test	K3	1	Lecture with Discussion and in class Assignment	Black Board
52	CO5	Examine the hypothesis concerning two means and their differences	K3	1	Lecture with Discussion and in class Assignment	Black Board
53	CO5	Examine the hypothesis concerning two proportions and their differences	K3	1	Lecture with Discussion and in class Assignment	Black Board
54	CO5	Examine the hypothesis concerning small samples using t-test (Single mean)	K3	1	Lecture with Discussion and in class Assignment	Black Board
55	CO5	Examine the hypothesis concerning small samples using t-test (Two means)	K3	1	Lecture with Discussion and in class Assignment	Black Board
56	CO5	Examine the hypothesis concerning variance using F-test	K3	1	Lecture with Discussion	Black Board

					and in class Assignment	
57	CO5	Examine the hypothesis concerning variance using F-test	K3	1	Lecture with Discussion and in class Assignment	Black Board
58	CO5	Examine the hypothesis using chi-square test (Goodness of fit)	K3	1	Lecture with Discussion and in class Assignment	Black Board
59	CO5	Examine the hypothesis using chi-square test (Goodness of fit) by using probability distributions	K3	1	Lecture with Discussion and in class Assignment	Black Board
60	CO5	Examine the hypothesis using chi-square test (Independence of attributes)	K3	1	Lecture with Discussion and in class Assignment	Black Board

Total No. of Classes: 60

Operating Systems

Academic Year: 2024-25

Year/ Semester: IV

Name of the Course: Operating Systems

Programme: B.Tech

Section: A,B,C& D

Course Code: V23CST06

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1	CO1	Describe Operating System Services, System Calls, Building, Booting and Debugging.	K2
2	CO2	Illustrate Process Management Concepts and CPU Scheduling Algorithms.	K3
3	CO3	Demonstrate Process Synchronization primitives and Deadlock handling techniques.	K3
4	CO4	Illustrate Memory Management Techniques and Page Replacement Algorithms.	K3
5	CO5	Describe File System Concepts and Mass Storage Structures.	K2

Text Books:

1. Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley, 2018.
2. Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson , 2016

REFERENCES

1. Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018
2. Operating Systems: A Concept Based Approach, D.M Dhamdhare, 3rd Edition, McGraw- Hill, 2013

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan

S. NO	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	1	-	
		Introduction of OS, Operating-System functions	K2	1	Lecture with Discussion	BB/ICT
2		Operating-System Operations, Computing environments	K2	1	Lecture with Discussion	BB/ICT
3		Free and Open-Source Operating Systems	K2	1	Lecture with Discussion	BB/ICT
4		Operating System Services, User and Operating-System Interface	K2	2	Lecture with Discussion	BB/ICT
5		System Calls, types of system calls	K2	1	Lecture with Discussion	BB/ICT
6		System programs, operating system design and implementation	K2	1	Lecture with Discussion	BB/ICT
7		Operating system Structure	K2	1	Lecture with Discussion	BB/ICT
8		Building and Booting an Operating System, Operating system debugging	K2	1	Lecture with Discussion	BB/ICT
		Total		9		

S. NO	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO2	Process Concept and Process Scheduling	K3	2	Lecture with Discussion	BB/ICT
2		Operations On Processes	K3	1	Lecture with Discussion	BB/ICT
3		Inter Process Communication	K3	1	Lecture with Discussion	BB/ICT
4		Threads overview	K3	1	Lecture with Discussion	BB/ICT
5		Multithreading Models, Thread libraries, issues	K3	1	Lecture with Discussion	BB/ICT
6		CPU Scheduling: Basic Concepts and CPU Scheduling Criteria	K3	1	Lecture with Discussion	BB/ICT
7		CPU Scheduling Algorithms	K3	3	Lecture with Discussion and in class Assignment	BB/ICT
8		Multiple processor Scheduling	K3	1	Lecture with Discussion	BB/ICT
		Total		11		

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO3	Critical Section Problem	K3	1	Lecture with Discussion	BB/ICT
2		Peterson's Solution	K3	1	Lecture with Discussion	BB/ICT
3		Mutex Locks	K3	1	Lecture with Discussion	BB/ICT
4		Semaphores	K3	1	Lecture with Discussion	BB/ICT
5		Monitors	K3	1	Lecture with Discussion	BB/ICT
6		Classic Problems of Synchronization	K3	2	Lecture with Discussion	BB/ICT
7		System Model and Dead Lock Characterization	K3	1	Lecture with Discussion	BB/ICT
8		Methods for Handling Deadlocks	K3	1	Lecture with Discussion	BB/ICT
9		Deadlock Prevention	K3	1	Lecture with Discussion	BB/ICT
10		Deadlock Avoidance	K3	1	Lecture with Discussion	BB/ICT
11		Deadlock Detection	K3	1	Lecture with Discussion and in class Assignment	BB/ICT
12		Recovery from Deadlock	K3	1	Lecture with Discussion	BB/ICT
		Total		13		

S. NO.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO4	Introduction, Contiguous Memory Allocation	K3	1	Lecture with Discussion	BB/ICT
3		Paging	K3	1	Lecture with Discussion	BB/ICT
4		Structure of the page table	K3	2	Lecture with Discussion and in class Assignment	BB/ICT
5		Swapping	K3	1	Lecture with Discussion	BB/ICT
6		Virtual memory introduction, Demand Paging, copy on write	K3	1	Lecture with Discussion	BB/ICT
7		Page Replacement Algorithms	K3	2	Lecture with Discussion and in class Assignment	BB/ICT
8		Allocation of Frames	K3	1	Lecture with Discussion	BB/ICT
9		Thrashing	K3	1	Lecture with Discussion	BB/ICT
10		Overview of Mass-Storage Structure	K2	1	Lecture with Discussion	BB/ICT
11		HDD scheduling	K2	1	Lecture with Discussion	BB/ICT
		Total		13		

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO5	File System interface: file concept, access methods	K2	1	Lecture with Discussion	BB/ICT
2		Directory structure, file system	K2	2	Lecture with Discussion	BB/ICT
3		File-system structure, File-system Operations	K2	1	Lecture with Discussion	BB/ICT
4		Directory implementation, Allocation method, Free space management	K2	1	Lecture with Discussion	BB/ICT
5		File-System Internals: File-System Mounting	K2	1	Lecture with Discussion	BB/ICT
6		Partitions and Mounting, File Sharing	K2	1	Lecture with Discussion	BB/ICT
7		Goals of protection, Principles of protection	K2	1	Lecture with Discussion	BB/ICT
8		Protection Rings, Domain of protection, Access matrix.	K2	1	Lecture with Discussion	BB/ICT
		Total		9		

Total No. of Classes: 55

Database Management Systems

Academic Year: 2024-25

Year/ Semester: IV

Name of the Course: Database Management Systems

Programme: B.Tech

Section: A,B,C & D

Course Code: V23CST07

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1	CO1	Describe Database Systems, various Data models and Database architecture.	K2
2	CO2	Develop various real time applications using Relational algebra and Relational calculus.	K3
3	CO3	Illustrate Nested queries and different types of joins.	K3
4	CO4	Apply various Normalization techniques to refine schema.	K3
5	CO5	Explain Transaction management, Concurrency control and Database indexing technique	K2

Text Books:

1. Database Management Systems, Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition TATA McGraw Hill.
2. An Introduction to Database Systems, C.J Date, A.Kannan, S.J Swamynathan 8th Edition, Pearson Education.

Reference Books:

1. Database Systems-Design, Implementation and Management, Peter Rob & Carlos Coronel 7th Edition, Course Technology Inc.
2. Fundamentals of Database Systems, Ramez Elmasri, Shamkant B. Navathe , 7th Edition, Pearson Education.
3. Database Systems - The Complete Book, Hector Garcia- Molina, Jeffry D Ullman, Jennifer Widom, 2nd Edition, Pearson.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		65	65	60	60	65
Targeted level of Attainment	Level 3	65	60	60	60	65
	Level 2	60	55	55	55	60
	Level 1	55	50	50	50	55

Lecture Plan:

UNIT-I: An Overview of Database System and Database Design						
S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs, COs, Introduction on DBMS and its Applications	K1	1	Lecture	BB
2		Define Data, Database, Database Management System	K1	1	Lecture	IFB
3		Characteristics of File System Vs Database, Database users	K1	2	Lecture with Discussion	IFB
4		Describe the concepts of Schema,	K2	1	Lecture	IFB
5		Explain three tier schema architecture for data independence	K2	1	Lecture with Discussion	IFB
6		Describe the database system structure, Explain database Environment	K1	1	Lecture with Discussion	IFB
8		Distinguish Centralized and Client/Server Architecture for the database	K2	2	Lecture with Discussion	IFB
9		Introduction of ER model, representation of entities	K2	1	Lecture with Discussion	IFB
10		Attributes , Entity set, Relationship, Relationship set, Constraints of ER model	K1	2	Lecture	IFB
11		Explain subclass, super class, inheritance	K2	1	Lecture With Discussion	IFB
12		Explain specialization, generalization using ER diagrams	K2	2	Lecture With Discussion	IFB
		TOTAL		16		

UNIT-II: RELATIONAL MODEL, RELATIONAL ALGEBRA AND RELATIONAL CALCULUS

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Explain the introduction to relational model, concepts of domain, attribute, tuple, relation	K2	1	Lecture With Discussion	IFB
2		Describe the importance of Null values	K2	1	Lecture With Discussion	IFB
3		Explain Domain constraints	K3	1	Lecture With Discussion	IFB
4		Explain Key constraints	K3	1	Lecture With Discussion	IFB
5		Explain integrity constraints and their importance	K3	1	Lecture With Discussion	IFB
6		Explain relational algebra	K3	2	Lecture With Discussion	IFB
7		Explain Relational Calculus	K3	2	Lecture With Discussion	IFB
8		Database Schema, Datatypes	K3	1	Lecture With Discussion	IFB
9		Table definition, different DML operations	K3	2	Lecture With Discussion	IFB
		TOTAL		14		

UNIT-III: STRUCTURED QUERY LANGUAGE (SQL)						
S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain the Form of a Basic SQL Query	K2	1	Lecture with Demonstration	IFB
2		Illustrate different clauses of SQL querying	K3	1	Lecture with Demonstration	IFB
3		Arithmetic & Logical operations	K3	1	Lecture with Demonstration	IFB
4		SQL Functions (Date and Time, Numeric, String Conversions)	K3	2	Lecture with Demonstration	IFB
5		Creating Tables with relationships	K3	1	Lecture with Demonstration	IFB
6		Implementation of key and integrity constraints	K3	1	Lecture with Demonstration	IFB
7		Nested Queries, Sub queries	K3	2	Lecture with Demonstration	IFB
8		Grouping, Aggregation, Ordering	K3	1	Lecture With Discussion	IFB
9		Implementation of different types of Joins	K3	2	Lecture With Discussion	IFB
10		View(Updatable and non-updatable)	K3	1	Lecture With Discussion	IFB
11		Relational set operations	K3	1	Lecture With Discussion	IFB
		TOTAL		14		

UNIT-IV: QUERIES, CONSTRAINTS, TRIGGERS AND SCHEMA REFINEMENT(NORMALIZATION)						
S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Schema refinement or purpose of Normalization	K3	1	Lecture With Discussion	IFB
2		Concept of Functional Dependency, Normal Forms based on functional dependency	K3	1	Lecture With Discussion	IFB
3		Lossless join and dependency preserving decomposition	K3	2	Lecture With Discussion	IFB
4		(1NF, 2NF, 3NF) Normal Forms	K3	2	Lecture With Discussion & Seminar	IFB
5		Concept of Surrogate Key, Boyce Codd normal Form	K3	1	Lecture With Discussion	IFB
7		MVD, Fourth Normal Form	K3	1	Lecture With Discussion	IFB
8		Fifth Normal Form	K3	1	Lecture	IFB
		TOTAL		9		

UNIT-V: TRANSACTION & CONCURRENCY CONTROL, STORAGE AND INDEXING

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1		Transaction State, ACID Properties	K2	1	Lecture With Discussion	ICT
2	CO 5	Concurrent Executions, Serializability	K2	1	Lecture With Discussion	ICT
3		Recoverability, Implementation of Isolation, ``Testing for Serializability	K2	1	Lecture With Discussion	ICT
4		Lock , timestamp, optimistic based concurrency protocols	K2	2	Lecture With Discussion	ICT
5		Deadlocks, Failure Classification	K2	1	Lecture With Discussion	ICT
6		Storage, Recovery and Atomicity, Recovery Algorithm	K2	1	Lecture With Discussion	ICT
7		Introduction to Indexing Techniques, B+ Trees	K3	2	Lecture With Discussion	ICT
8		Operations on B+ Trees	K3	1	Lecture With Discussion	ICT
9		Hash Based Indexing	K3	2	Lecture With Discussion	ICT
		TOTAL		12		

Total No. of Classes : 65

Software Engineering

Academic Year: 2024-25

Year/ Semester: IV

Name of the Course: Software Engineering

Programme: B.Tech

Section: A,B,C & D

Course Code: V23CST08

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1	CO1	Demonstrate the evolution, development practices, and various software life cycle models. (K3)	K3
2	CO2	Illustrate project management techniques and explain requirements analysis and specification. [K3]	K3
3	CO3	Develop effective software designs, using agile methodologies and user interface principles. [K3]	K3
4	CO4	Apply coding practices, conduct thorough testing, and manage software quality. (K3)	K3
5	CO5	Distinguish the use of CASE tools, manage software maintenance, and implement software reuse Strategies. [K4]	K4

Text Books:

1. Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI.
2. Software Engineering A Practitioner's Approach, Roger S. Pressman, 9th edition, McGraw Hill International Edition.

REFERENCES

1. Software Engineering, Ian Sommerville, 10th Edition, Pearson.
2. Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	55	55	55	55	55
	Level 1	50	50	50	50	50

Lecture Plan:

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs, COs, Introduction: Identify need of Software Engineering	K1	1	Lecture	IFP/ICT
2.		Explain the importance of SE, Evolution, Software development projects.	K2	2	Lecture	IFP/ICT
3.		Describe the Exploratory style of software developments, Emergence of software engineering.	K2	2	Lecture+ discussion	IFP/ICT
4.		Explain the Notable changes in software development practices, Computer system engineering.	K2	2	Lecture	IFP/ICT
5.		Explain Basic concepts, Waterfall model and its extensions,	K2	2	Lecture + discussion	IFP/ICT
6.		Illustrate the Rapid application Development, Agile development model, Spiral model.	K3	2	Lecture + discussion	IFP/ICT
		TOTAL		11		

UNIT-II

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO2	Describe the Metrics for project size estimation, Project estimation techniques.	K2	2	Lecture	IFP/ICT
2.		Describe the project estimation Techniques.	K2	2	Lecture+ discussion	IFP/ICT
3.		Illustrate Empirical estimation techniques and COCOMO	K3	2	Lecture	IFP/ICT
4.		Demonstrate Halstead 's software science and risk management.	K3	2	Lecture + discussion	IFP/ICT
5.		Describe the requirements gathering and analysis, Software Requirements Specification (SRS)	K2	2	Lecture + discussion	IFP/ICT
6.		Explain Formal system specification, Axiomatic specification, Algebraic specification, Executable specification and 4GL.	K2	2	Lecture + discussion	IFP/ICT
		TOTAL		13		

UNIT-III

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO3	Describe Cohesion and Coupling, approaches to software design	K2	2	Lecture	IFP/ICT
2.		Discuss Agility and the Cost of Change, Agile Process, Extreme Programming (XP)	K2	2	Lecture+ discussion	IFP/ICT
3.		Outline Other Agile Process Models, Tool Set for the Agile Process	K1	1	Lecture	IFP/ICT
4.		Explain the Overview of SA/SD methodology, Structured analysis, DFD model of a system	K2	2	Lecture + discussion	IFP/ICT
5.		Demonstrate Structured design, Detailed design, and Design Review.	K2	1	Lecture + discussion	IFP/ICT
6.		Explain Characteristics of a good user interface, and types of user interfaces.	K2	2	Lecture	IFP/ICT
7.		Illustrate the fundamentals of component-based GUI development and UI design methodology.	K3	1	Lecture	IFP/ICT
		TOTAL		13		

UNIT-IV

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO4	Demonstrate, Black-box testing, White-Box testing, Debugging, Program analysis tools.	K3	2	Lecture	IFP/ICT
2.		Discuss Integration testing, Testing object-oriented programs, Smoke testing, and issues associated with testing.	K2	2	Lecture+ discussion	IFP/ICT
3.		Explain reliability of software, Statistical testing, Software quality.	K2	2	Lecture	IFP/ICT
4.		Discuss about Software quality management system, ISO 9000, SEI Capability maturity model	K2	2	Lecture + discussion	IFP/ICT
5.		Outline important quality standards, and Six Sigma.	K1	2	Lecture + discussion	IFP/ICT
		TOTAL		11		

UNIT-V

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO5	Discuss the characteristics of CASE tools, second generation CASE Tools, and Architecture of a CASE Environment.	K2	2	Lecture	IFP/ICT
2.		Explain Characteristics of software maintenance, Software reverse engineering	K2	2	Lecture+ discussion	IFP/ICT
3.		Illustrate the Software Maintenance process models and Estimation of maintenance cost.	K4	2	Lecture	IFP/ICT
4.		Explain reuse- definition, introduction, reason behind no reuse so far	K2	2	Lecture + discussion	IFP/ICT
5.		Demonstrate the Basic issues in any reuse Program, reuse approach and Reuse at organization level.	K3	2	Lecture + discussion	IFP/ICT
		TOTAL		12		

Total No. of Classes: 60

Operating Systems Lab

Academic Year: 2024-25

Year/ Semester: IV

Name of the Course: Operating Systems Lab

Programme: B.Tech

Section: A,B , C & D

Course Code: V23CSL06

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO. No.	Course Outcome	BTL
1	CO1	Practice of UNIX fundamentals, commands & system calls. [K3]	K3
2	CO2	Construct programs using CPU scheduling algorithms.[K3]	K3
3	CO3	Demonstrate IPC, semaphores, monitors, deadlocks. [K3]	K3
4	CO4	Develop programs to solve Page replacement algorithms, file allocation strategies and memory allocation strategies [K3]	K3

Reference Books:

1. Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley, 2018.
2. Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson, 2016
3. Operating Systems -Internals and Design Principles, Stallings W, 9th edition,Pearson,2018.
4. Operating Systems: A Concept Based Approach, D.M Dhamdhere, 3rd Edition, McGraw- Hill, 2013

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		75	75	75	75	75
Targeted level of Attainment	Level 3	70	70	70	70	70
	Level 2	65	65	65	65	65
	Level 1	55	55	55	55	55

Lecture Plan:

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Practice Basic UNIX Commands.	K3	3	Lecture With Discussion	ICT
2		Develop programs using the following UNIX operating system calls fork, exec, getpid, exit, wait, close, stat, opendir and readdir	K3	3	Lecture With Discussion	ICT
3		Demonstrate UNIX commands like cp, ls, grep, etc.,	K3	3	Lecture With Discussion	ICT
			Total Hours	9		

S. No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Demonstrate the following CPU scheduling algorithms a) FCFS b) SJF c) Priority d) Round Robin	K3	6	Lecture With Discussion	PPT
2		Illustrate concurrent execution of threads using pthreads library.	K3	3	Lecture With Discussion	PPT
			Total Hours	9		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Develop a program to control the number of ports opened by the operating system with a) Semaphore b) Monitors.	K3	3	Lecture with Demonstration	PPT
2		Solve producer-consumer problem using Semaphores	K3	3	Lecture with Demonstration	PPT
3		Demonstrate Bankers Algorithm for Dead Lock avoidance and prevention.	K3	3	Lecture with Demonstration	PPT
			TOTAL HOURS	9		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Demonstrate the following memory allocation methods for fixed partition a) First fit b) Worst fit c) Best fit	K3	3	Lecture with Discussion	ICT
2		Develop programs to simulate the following page replacement algorithms a) FIFO b) LRU c) LFU	K3	3	Lecture with Discussion	ICT
3		Develop program to simulate Paging Technique of memory management.	K3	3	Lecture with Discussion	ICT
			Total Hours	9		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Develop program to simulate the following file allocation strategies a) Sequential b) Indexed c) Linked	K3	3	Lecture with Discussion	PPT
2		Download and install nachos operating system and experiment with it.	K3	3	Lecture with Discussion	PPT
		TOTAL		6		

Total No. of Classes: 42

Database Management System Lab

Academic Year: 2024-25

Programme: B.Tech

Year/ Semester: IV

Section: A,B,C,D

Name of the Course: Database Management System Lab

Course Code: V23CSL07

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1	CO1	Construct SQL Queries using DDL, DML and DCL commands.	K3
2	CO2	Experiment with nested queries, built-in functions and various constraints.	K3
3	CO3	Construct PL/SQL programs on Control structures, Cursors and Triggers.	K3
4	CO4	Develop a Java program to connect with a database using JDBC.	K3

Text Books:

1. Oracle Database 11g the Complete Reference by Oracle Press, Kevin Loney
2. Database Systems Using Oracle, Nilesh Shah, 2nd Edition ,PHI.
3. Introduction to SQL, Rick F Vander Lans, 4th Edition, Pearson Education.

Reference Books:

1. Oracle PL/SQL Interactive Workbook, B. Rosenzweig and E. Silvestrova, 2nd Edition, Pearson education.
2. SQL & PL/SQL for Oracle 10 g, Black Book, Dr. P. S. Deshpande, Dream Tech.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		75	75	75	70	70
Targeted level of Attainment	Level 3	75	75	75	70	70
	Level 2	65	65	65	60	60
	Level 1	55	55	55	50	50

Lecture Plan

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Exp1:.. Demonstrate Creation, altering and dropping of tables and inserting rows into a table, SELECT command	K3	9	Demonstration & Experiment	ICT
2		Exp2: Construct queries using ANY, ALL, In, EXISTS, NOT EXISTS, UNION, INTERSECT, Constraints				
3		Exp3: Develop different queries using Group By, Order By, and Having Clauses and Aggregate Functions				
4		Exp4:Develop queries using Conversion functions, Date, String fuctions				

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
5	CO2	Exp5: Develop a simple PL/SQL program , Exception Handling. COMMIT, ROLLBACK AND SAVEPOINT in PL/SQL	K3	12	Lecture & Experiment	ICT
6		Exp6: Develop PL/SQL program using NESTED IF, CASE AND CASE expression. Programs using NULLIF and COALESCE functions.				
7		Exp 7:.. Develop PL/SQL program using WHILE LOOPS, FOR LOOPS, ERROR handling , BUILT-IN Exceptions, USER defined exceptions				

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
10 11 12	CO3	Exp8:Develop programs using procedures, IN, OUT parameters. Exp 9: Develop programs on stored functions Exp 10 : Develop programs using CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables Exp 11: Develop programs using BEFORE and AFTER Triggers. Exp 12 : Construct a table and perform search using indexing and non-indexing.	K3	12	Lecture & Experiment	ICT

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
13 14 15	CO4	Exp13: Develop a Java program that connects to a database using JDBC. Exp14: Develop a Java program to connect to a database using JDBC and insert values into it. Exp15: : Develop a Java program to connect to a database using JDBC and delete values from it.	K3	9	Demonstration & Experiment	ICT

Total No. of Classes:: 42

Full Stack Development-I

Academic Year: 2024-25

Year/ Semester: IV

Name of the Course: Full Stack Development-I

Programme: B.Tech

Course Code: V23CSSE02

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1	CO1	Illustrate HTML elements and their attributes for designing static web pages.	K3
2	CO2	Apply appropriate CSS styles to HTML elements.	K3
3	CO3	Demonstrate JavaScript Pre-defined and User-Defined Objects.	K3
4	CO4	Develop dynamic web pages and validate forms using JavaScript.	K3

Reference Books:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Web Programming with HTML5, CSS and JavaScript, John Dean, Jones & Bartlett Learning, 2019 (Chapters 1-11).
3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, A Press, O'Reilly.

Online Learning Resources:

1. <https://www.w3schools.com/html>
2. <https://www.w3schools.com/css>
3. <https://www.w3schools.com/js/>
4. <https://www.w3schools.com/nodejs>

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4
Targeted Proficiency Level		70	70	70	70
Targeted level of Attainment	Level 3	70	70	70	70
	Level 2	60	60	60	60
	Level 1	50	50	50	50

Lecture Plan

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.		Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	-
2.	CO1	Develop a HTML program on Lists	K3	2	Demonstration & Experiment	ICT
3.		Develop a HTML program on Links	K3	2	Demonstration & Experiment	ICT
4.		Develop a HTML program on Images	K3	2	Demonstration & Experiment	ICT
5.		Develop a HTML program on Tables	K3	3	Demonstration & Experiment	ICT
6.		Develop a HTML program on Forms	K3	3	Demonstration & Experiment	ICT
7.		Develop a HTML program on Frames	K3	3	Demonstration & Experiment	ICT

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO2	Develop a HTML program using CSS	K3	3	Demonstration & Experiment	ICT
2.		Develop a HTML program using Selectors forms (element, id, class, group, universal)	K3	3	Demonstration & Experiment	ICT
3.		Develop a HTML program using CSS with Color options.	K3	3	Demonstration & Experiment	ICT
4.		Develop a HTML program using Font and Text options	K3	3	Demonstration & Experiment	ICT
5.		Develop a HTML program using CSS Box Model	K3	3	Demonstration & Experiment	ICT

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO3	Develop a JavaScript program to embed Internal/External JavaScript in webpage	K3	4	Demonstration & Experiment	ICT
2.		Develop a JavaScript program which prompt dialog box to read Input and display Output	K3	3	Demonstration & Experiment	ICT
3.		Develop a JavaScript program using document object properties and methods.	K3	5	Demonstration & Experiment	ICT
4.		Develop a JavaScript program using conditional Statements	K3	3	Demonstration & Experiment	ICT
5.		Develop a JavaScript program using loops	K3	3	Demonstration & Experiment	ICT

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO4	Develop a JavaScript program using functions	K3	4	Demonstration & Experiment	ICT
2.		Develop a dynamic webpage JavaScript program using event handling mechanism .	K3	4	Demonstration & Experiment	ICT
3.		Develop a JavaScript program to validate user input	K3	4	Demonstration & Experiment	ICT

Total Number of Classes:60

Environmental Studies

Academic Year: 2024-25

Programme: B.Tech

Year/ Semester: IV

Course Code: V23CEACO1

Name of the Course: Environmental Studies

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1	CO1	To make the students to get awareness on environment.	K2
2	CO2	To understand the importance of protecting natural resources, ecosystems for future generations and pollution causes due to the day to day activities of human life.	K3
3	CO3	To save earth from the inventions by the engineers.	K3

Text Books/ Reference Books:

1. Text book of Environmental Studies by : Dr. KVSG. Murali Krishna, VGS Publishers, Vijayawada.
2. Text book of Environmental Studies by R. Rajagopalan 2nd edition, 2011, Oxford University Press.
3. Text book of Environmental Studies by: Palanisamy et al., Pearson Education, Chennai.
4. Text book of Environmental Studies by Shaashi Chawla, TMH, NewDelhi.
5. Text book of Environmental Studies – MahuaBasu& Xavier, Cambridge University Press.
6. Text book of Environmental Studies by: Benny Joseph, TMH Co, New Delhi.
7. Text book of Environmental Studies by: Deeshita Dave & P. UdayaBhaskar, Cengage Learning.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		75	70	70	70	70
Targeted level of Attainment	Level 3	65	60	65	75	60
	Level 2	60	60	60	60	60
	Level 1	50	50	50	50	50

Lecture Plan

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO1	UNIT-1	K2	10	Integrated Teaching	IFP
2.		Multi-disciplinary nature of the environmental studies.		1	Integrated Teaching	
3.		Definition, Scope & importance, need for public awareness.		1	Integrated Teaching	
4.		Natural resources and associated problems.		1	Integrated Teaching	
5.		Forest resources-use and overexploitation ,deforestation.		1	Integrated Teaching	Charts
6.		Timber extraction –mining ,dams, effects on forest and tribal people.		1	Integrated Teaching	IFP
7.		Water resources- use and overutilization of surface and ground water		1	Integrated Teaching	
8.		Floods ,drought, conflicts over water, dams-benefits and problems.		1	Integrated Teaching	
9.		Mineral resources-use and over exploitation.		1	Lecture with discussion	Video Lecture by Projector
10.		Food resources-world food problems ,effects of modern agriculture.		1	Lecture with discussion	IFP
11.		Energy resources		1	Integrated Teaching	

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
12.	CO2	UNIT-2	K3	09	Lecture with discussion	IFP
13.		Ecosystem-concept –structure and function of an ecosystem.		1	Lecture with discussion	
14.		Energy flow in the ecosystem-ecological succession.		1	Lecture with discussion	
15.		Food chains ,food webs & ecological pyramids.		1	Lecture with discussion	
16.		Forest ecosystem, grassland ecosystem.		1	Lecture with discussion	
17.		Desert ecosystem, aquatic ecosystem.		1	Lecture with discussion	Video Lecture by Projector
18.		Biodiversity-definition ,levels ,biogeographical classification of india		1	Lecture with discussion	
19.		Values of biodiversity		1	Lecture with discussion	IFP
20.		Indian biodiversity, Hot spots, Endangered species		1	Integrated Teaching	
21.		Threats & conservation		1	Integrated Teaching	

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
22.	CO3	UNIT-3: ENVIRONMENTAL POLLUTION	K4	07		IFP
23.		Definition, causes, effects & control measures of Air pollution.		1	Integrated Teaching	
24.		Water pollution		1	Integrated Teaching	
25.		Soil Pollution		1	Integrated Teaching	
26.		Noise pollution, Marine pollution.		1	Integrated Teaching	
27.		Thermal pollution, nuclear hazards		1	Integrated Teaching	
28.		Solid waste management		1	Integrated Teaching	
29.		Disaster management		1	Integrated Teaching	

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
30.	CO3	UNIT-3: ENVIRONMENTAL POLLUTION	K4	07		IFP
31.		Definition, causes, effects & control measures of Air pollution.		1	Integrated Teaching	
32.		Water pollution		1	Integrated Teaching	
33.		Soil Pollution		1	Integrated Teaching	
34.		Noise pollution, Marine pollution.		1	Integrated Teaching	
35.		Thermal pollution, nuclear hazards		1	Integrated Teaching	
36.		Solid waste management		1	Integrated Teaching	
37.		Disaster management		1	Integrated Teaching	

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
38.	CO4	UNIT-4: SOCIAL ISSUES AND THE ENVIRONMENT.	K3	08		IFP
39.		Unsustainable to sustainable development-		1		
40.		Urbanization, Water conservation.		1		
41.		Resettlement & Rehabilitation Issues		1		
42.		Environmental Ethics ,waste land reclamation.		1		
43.		Climatic changes-global warming, acid rains, ozone layer depletion.		1		
44.		Environmental Protection Act, Water pollution Act.		1		
45.		Air pollution Act ,Wild life Act.		1		
46.		Forest conservation Act- Environmental Legislation, public awareness.		1		

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
47.	CO5	UNIT-5: HUMAN POPULATION AND THE ENVIRONMENT	K2	06	Lecture with discussion	IFP
48.		Population growth, variation among nations.		1	Integrated Teaching	
49.		Family welfare programmes-environment and human health.		1	Integrated Teaching	
50.		Human rights-value education		1	Integrated Teaching	
51.		HIV /AIDS-Women and child welfare.		1	Integrated Teaching	
52.		Role of IT in environment and human health.		1	Integrated Teaching	
53.		Field Visit		1	Integrated Teaching	

Total Number of Classes:40

Professional Communication Skills - II

Academic Year: 2024-25

Programme: B.Tech.

Year/Sem: IV

Course Code: V20ENT02

Name of the Course: Professional Communication Skills - II

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S. No.	CO No.	Course Outcome	BTL
1	CO1	Define the meaning and significance of Soft Skills in contrast to Hard Skills and English, and learn how to enhance them. (K1)	K1
2	CO2	Identify SWOT and demonstrate soft skills competently in the workplace through intrapersonal skills. (K2)	K2
3	CO3	Demonstrate interpersonal skills by displaying leadership and teamwork in the working environment for professional excellence. (K3)	K3
4	CO4	Employ oral communication skills effectively and have an approachable demeanour in the workplace. (K3)	K3
5	CO5	Produce literacy skills effortlessly, meeting the requirements of the corporate world. (K3)	K3

Text Books:

1. Sherfield, M. Robert et al, Cornerstone Developing Soft Skills, 4/e, Pearson Publication, New Delhi, 2014.
2. Alka Wadkar, Life Skills for Success, 1/e, Sage Publications India Private Limited, 2016.

Reference Books :

1. Sambaiah.M. Technical English, Wiley publishers India. New Delhi. 2014.
2. Gangadhar Joshi, From Campus to Corporate, SAGE TEXT.
3. Alex.K, Soft Skills, 3rd ed. S. Chand Publication, New Delhi, 2014.
4. Meenakshi Raman and Sangita Sharma, Technical Communication: Principle and Practice, Oxford University Press, 2009.
5. Shalini Varma, Body Language for Your Success Mantra, 4/e, S. Chand Publication, New Delhi, 2014.
6. Stephen Covey, Seven Habits of Highly Effective People, JMD Book, 2013.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		55	55	55	55	55
Targeted level of Attainment	Level 3	55	55	55	55	55
	Level 2	50	50	50	50	50
	Level 1	45	45	45	45	45

S. No	COS	Intended Learning Outcomes (ILO)	Knowledge Level	No. of Hours	Pedagogy	Teaching aids
1	CO1	Unit-I: Introduction Define the meaning and describe the importance of Soft Skills.	K1	1	Lecture	IFP
2		Identify the difference between Soft Skills and Hard Skills.	K1	1	Lecture with activity	IFP & Handout
3		Find the similarities and differences between Soft Skills and English.	K1	1	Lecture with activity	IFP & Handout
4		Present the techniques of Soft Skills to enhance the quality of professional life.	K1	1	Lecture with activity	IFP
4						
5	CO2	Unit-II : Intra-Personal Infer the meaning and introspect SWOT to get ready for the corporate world.	K2	2	Lecture with Discussion and Activity	IFP & Handout

6		Identify the ways to reach the goal via the acquaintance of all other intrapersonal skills.	K2	1	Lecture with Discussion and Activity	IFP & Handout
7		Interpret how Emotional Intelligence and Attitudinal Skills can improve their relationship and enhance overall well-being.	K2	2	Lecture with Discussion and Activity	IFP & Handout
8		Illustrate real-life scenarios from multiple perspectives with their critical thinking and analytical problem-solving skills.	K2	2	Lecture with Discussion and Activity	IFP & Handout
9		Clarify and prioritize tasks, manage time to meet deadlines, and balance personal and professional life.	K2	1	Lecture with Discussion and Activity	IFP & Handout
8						
10	CO3	Unit-III: Inter-Personal Employ presentation skills, public speaking, and confrontational conversation with a client effectively.	K3	2	Lecture with Discussion and Activity	IFP & Handout
11		Differentiate between group and team and how teamwork is crucial in the workplace to increase productivity.	K2	1	Lecture with Discussion and Activity	IFP & Handout

12		Show managerial and leadership skills to plan, delegate, and evaluate tasks and projects.	K3	1	Lecture with Discussion and Activity	IFP
13		Demonstrate persuasive and negotiation skills to handle objections and to maintain good relationships with one and all in the corporate world.	K3	1	Lecture with Discussion and Activity	IFP
5						
14	CO4	Unit-IV: Oral Communication Skills (L&S) & Kinesics Apply active listening to deduce potential distractions and approach conversation openly, avoiding preconceived notions.	K3	4(2+2)	Lecture with Discussion and Activity	IFP
15		Relate nonverbal communication to convey emotions and attitudes across different cultures and contexts.	K3	1	Lecture with Discussion and Activity	IFP & Handout
16		Show passion in mock interviews for an aspiring role in the corporate sector by speaking clearly and accurately at a moderate pace.	K3	4(2+2)	Lecture with Discussion and Activity	IFP
9						

17	CO5	Unit-V : Literacy Skills(R&W) Find the difference between literal and inferential reading Comprehension and identify the central theme to draw conclusions.	K3	4(2+2)	Lecture with Discussion and Activity	IFP & Handout
18		Prepare a structural, technical report that provides clear information and research and experimental results.	K3	1	Lecture with Discussion and Activity	IFP
19		Construct a resume with a brief overview of their education, skills, achievements, and awards by including a well-written cover letter.	K3	1	Lecture with Discussion and Activity	IFP
6						

Total No. of Classes: 44